Ch6.2 Regularization

Thursday, October 15, 2026 9:24 AM

Recap: Battle overfitting

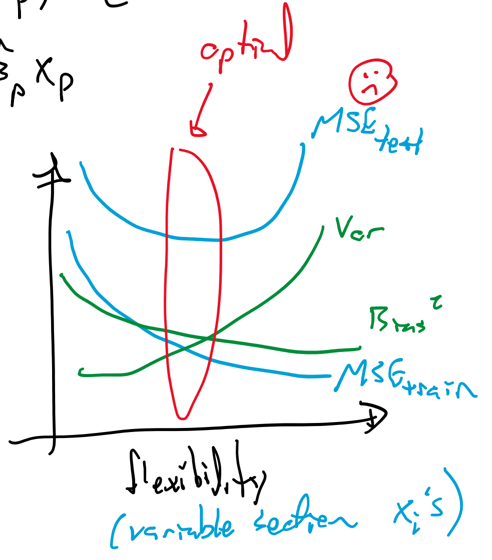
LR $y = f(x) + \varepsilon = (\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p) + \varepsilon$

$f(x) \approx \hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p$

2 goals WRT LR

① $p \approx n$, avoid overfitting →

✪ ② If p largeish (20), hard to interpret $\hat{\beta}_i$'s.



2 approaches: ① Variable selection by trying many x_i combinations (best, $\underbrace{\text{forw, back}}_{\text{practical}}$)

& assess via

(i) stat metrics (adj R^2, BIC, AIC, C_p)

✪ (ii) CV

② Regularization: when fit $\hat{f}$, include variable selection by restricting $\hat{\beta}_i$'s.

shape of cost function

only shrinkage

(i) Ridge (ℓ^2-penalty)

(ii) LASSO (ℓ^1-penalty)

↳ least absolute shrinkage & $\boxed{\text{selection}^{*}}$ operator

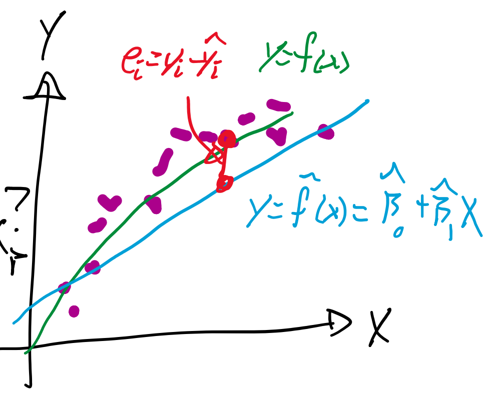
combo

(iii) Elastic net (lab)

Recap: OLS (ordinary least squares)

$Y = f(x) + \varepsilon = (\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p) + \varepsilon$

How to fit $f(x) \approx \hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p$?



values of $\vec{\beta}$ at which

$$\hat{\vec{\beta}} = \underset{\vec{\beta}}{\boxed{\text{arg}}\,\text{min}}\ RSS = \underset{\vec{\beta}}{\text{arg min}} \sum_{i=1}^{n} e_i^2$$

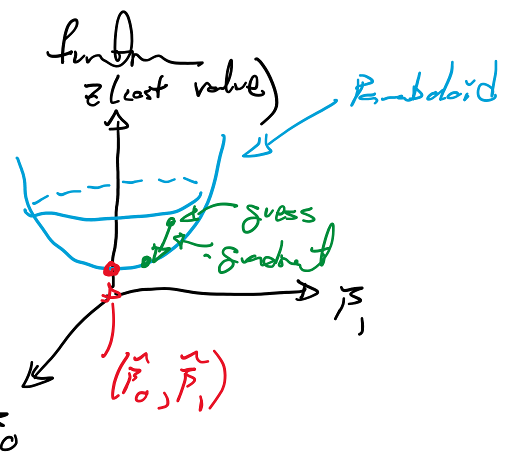
$$= \underset{\vec{\beta}}{\text{arg min}} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$= \underset{\vec{\beta}}{\text{arg min}} \underbrace{\sum_{i=1}^{n} \Big(y_i - \big(\underbrace{\beta_0 + \sum_{j=1}^{p} \beta_j X_{ji}}_{f(x)}\big)\Big)^2}_{\text{Cost function}}$$

(2 ways to optimize: ① Normal eqns ② $\underbrace{\text{Gradient descent}}_{\text{Genl ML}}$)

G.D. Plot of cost function

SLR



Details in NN.

Ridge Regression Idea: When OLR overfits, $\hat{\beta}_i$'s get very big. Force them to stay small (& not overfit).

Standardize (z-score) $z_i = \dfrac{x_i - \bar{x}_i}{SD(x_i)}$ (x_i = orig) (mean 0 & SD 1)

Modify cost function

regularization param to tune

β_j gets big, adds to RSS & optimization avoids

$$\hat{\vec{\beta}}^{R}_{\lambda} = \underset{\vec{\beta}}{\text{arg min}} \Big(RSS + \underbrace{\lambda \sum_{j=1}^{p} \beta_j^2}_{\vec{\beta}\text{ penalty}} \Big)$$

shrinkage

Warning:

OLS gives same $\vec{\beta}$'s regardless of data scale (need not normalize)

Ridge regression needs normalization else $\hat{\beta}^{R}_{i}$ skew per x_i units (un-normalize after fitting)

$$= \underset{\vec{\beta}}{\text{arg min}} \Big(\sum_{i=1}^{n} \big(y_i - (\beta_0 + \sum_{i=1}^{p} \beta_i X_i)\big)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \Big)$$

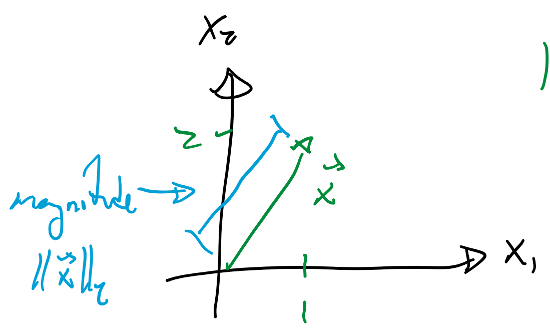
(via G.D.)

Recall: $\sum_{i=1}^{p} \beta_i^2 = \|\vec{\beta}\|_2^2$, $\vec{\beta} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$

ℓ^2-norm

Euclidean dist

$\|\vec{x}\|_2 = \sqrt{1^2 + 2^2} = \sqrt{5}$



Tune λ to balance over/under fitting:

CV on λ range!

- $\lambda \approx 0$, OLS ish (overfit)
- ★ between best
- λ big, $\hat{\beta}_i$'s super small (underfit)

$\vec{\beta}$

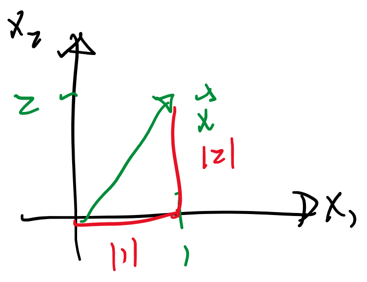
$\hat{\vec{\beta}}^{R}_{\lambda}$

LASSO Same as ridge regr, but $\|\cdot\|_1$ (ℓ^1-norm instead).

$$\hat{\vec{\beta}}^{L}_{\lambda} = \underset{\vec{\beta}}{\text{arg min}} \Big(RSS + \lambda \sum_{j=1}^{p} |\beta_j| \Big)$$

(Difference is β_j's get 0ed! cut & shrunk)

Note: $\|\vec{x}\|_1 = |1| + |2| = 3$ (taxicab or manhattan distance)



Note: Optimization reform. gives LASSO intuition.